THE COMPLETE GRAPH MAXIMIZES THE VARIANCE OF THE ANTIVOTER MODEL

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Abstract. In the antivoter model on a finite graph each vertex, at rate one, adopts the opinion opposite to that of a uniformly chosen neighbour. Let S be the sum of the ± 1 opinions in stationarity. Aldous and Fill asked whether, among regular graphs on n vertices, $\operatorname{Var} S$ is largest for the complete graph, where it equals $n(n-2)/(2n-3)$. We prove that it is: for every connected non-bipartite regular graph on $n \geq 3$ vertices and every stationary distribution, $\operatorname{Var} S \leq n(n-2)/(2n-3)$, with equality only for K_n. The proof writes the pair correlations as the solution of a Sylvester equation and reduces the question to a lower bound on the diagonal of $(I+P)^{-1}$, where P is the transition matrix of simple random walk. For vertex-transitive graphs the same argument gives $\operatorname{Var} S = n^2/\big(2\sum_i (1+\lambda_i)^{-1}\big)$, the sum running over the eigenvalues of P.

1. Introduction

Let $G = (V, E)$ be a finite connected d-regular graph on $n \geq 3$ vertices. The antivoter model on G is the continuous-time Markov chain on $\{-1, 1\}^V$ in which each vertex v, at the times of a rate-one Poisson clock, chooses a neighbour u uniformly at random and sets $\eta_v := -\eta_u$. On finite graphs it was studied by Donnelly and Welsh [2], who showed that when G is neither bipartite nor a cycle the chain is irreducible on the $2^n - 2$ non-constant configurations; on a bipartite graph it is absorbed in one of the two proper 2-colourings. Write $S = \sum_v \eta_v$ for the opinion margin under a stationary distribution. The antivoter model is a standard test case for normal approximation by Stein's method [3, 4], where $\operatorname{Var} S$ enters as the normalising constant.

Section 14.4 of Aldous and Fill [1] develops the variance theory: $\mathbb{E}S = 0$, $\operatorname{Var} S < n/2$ for every such graph (their Proposition 14.22), $2\kappa/d \leq \operatorname{Var} S$ with κ a cut parameter (Corollary 14.21), and for the complete graph the correlation of two distinct vertices is $-1/(2n-3)$, so that

$$\operatorname{Var} S = \frac{n(n-2)}{2n-3} \qquad (G = K_n). \tag{1.1}$$

They then ask for a proof that $\operatorname{Var} S$ is maximized by the complete graph, and remark that this might follow from a sharpening of the inequality behind $\operatorname{Var} S < n/2$. We answer the question.

Theorem 1.1. *Let G be a connected non-bipartite d-regular graph on $n \geq 3$ vertices, and let η have any stationary distribution of the antivoter model on G. Then*

$$\operatorname{Var} S \leq \frac{n(n-2)}{2n-3},$$

with equality if and only if $G = K_n$.

The hypothesis that G is not a cycle, needed for irreducibility, plays no role: the bound holds for every stationary distribution, and the odd cycles are covered. Bipartite graphs are excluded because there the stationary distributions are mixtures of the two proper colourings, on which $S = \pm(|A| - |B|)$ for the colour classes A, B, so $\operatorname{Var} S$ can be of order n^2.

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The argument also yields an exact formula for vertex-transitive graphs. Let $P = A_G/d$ be the transition matrix of simple random walk, with eigenvalues $1 = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_n > -1$.

Corollary 1.2. *If G is in addition vertex-transitive, then*

$$\operatorname{Var} S = \frac{n^2}{2\sum_{i=1}^{n}(1+\lambda_i)^{-1}}.$$

For K_n the eigenvalues are 1 and $-1/(n-1)$ with multiplicity $n-1$, and the corollary returns (1.1). For the octahedron $K_{2,2,2}$ ($\lambda = 1, 0, 0, 0, -\frac{1}{2}, -\frac{1}{2}$) it gives $36/15 = 2.4$, and for the prism $C_3 \times K_2$ ($\lambda = 1, \frac{1}{3}, 0, 0, -\frac{2}{3}, -\frac{2}{3}$) it gives $36/18.5 \approx 1.9459$; both agree with the exact solution of the correlation equations in Table 1.

2. The correlation equations

Fix a stationary distribution and put $c(v,w) = \mathbb{E}[\eta_v \eta_w]$, so $c(v,v) = 1$, and let $C = (c(v,w))_{v,w\in V}$. Let $Q = I + P$. Since G is connected and not bipartite, the eigenvalues of P lie in $(-1, 1]$, so Q is symmetric and positive definite with $Q\mathbf{1} = 2\mathbf{1}$.

Lemma 2.1. *$\mathbb{E}\eta_v = 0$ for every v, and there are numbers $\delta_v \geq 0$ such that*

$$QC + CQ = D, \qquad D = \operatorname{diag}(\delta_v), \qquad \delta_v = 2\Big(1 + \frac{1}{d}\sum_{u\sim v} c(u,v)\Big). \tag{2.1}$$

Moreover $\operatorname{Var} S = \frac{1}{4}\sum_v \delta_v$.

Proof. Under stationarity the generator applied to any function has mean zero. For $f(\eta) = \eta_v$ only the clock at v changes f, and it replaces η_v by $-\eta_u$ with u a uniform neighbour, so $0 = -\mathbb{E}\eta_v - (Pm)_v$ with $m_v = \mathbb{E}\eta_v$. Thus $Qm = 0$ and $m = 0$. For $f(\eta) = \eta_v\eta_w$ with $v \neq w$, the clocks at v and at w each change f, and

$$0 = -2c(v,w) - \sum_u P(v,u)c(u,w) - \sum_u P(w,u)c(v,u),$$

which is the (v,w) entry of $QC + CQ = 0$ off the diagonal (the term $u = w$ in the first sum is $c(w,w) = 1$, as it should be, since $\eta_w\eta_w = 1$). Hence $QC + CQ$ is diagonal; its (v,v) entry is $2(QC)_{vv} = 2(1 + (PC)_{vv})$, which is the δ_v of (2.1), and $\delta_v \geq 0$ because $|c| \leq 1$. Finally, $\mathbb{E}S = 0$ by the first part, and multiplying (2.1) by $\mathbf{1}$ on both sides gives $4\,\mathbf{1}^\top C\mathbf{1} = \sum_v \delta_v$, since $Q\mathbf{1} = 2\mathbf{1}$ and Q is symmetric; and $\mathbf{1}^\top C\mathbf{1} = \mathbb{E}S^2 = \operatorname{Var} S$. □

In terms of correlations, Lemma 2.1 says $\operatorname{Var} S = \frac{n}{2}(1 + \bar{c})$, where $\bar{c}$ is the average of $c(u,v)$ over the $nd/2$ edges; this is Lemma 14.20 of [1] in another form, and Corollary 14.24 there when G is edge-transitive.

Lemma 2.2. *With $m_u = (Q^{-1})_{uu}$,*

$$\sum_{u\in V} \delta_u\, m_u = 2n.$$

Proof. The linear map $X \mapsto QX + XQ$ on symmetric matrices has eigenvalues $(1+\lambda_i)+(1+\lambda_j) > 0$, so it is invertible and (2.1) determines C from D. Since $e^{-tQ} \to 0$ exponentially fast, the matrix $\int_0^\infty e^{-tQ} D e^{-tQ}\, dt$ is well defined, and it solves (2.1), because $Q(\cdot) + (\cdot)Q$ applied to the integrand is $-\frac{d}{dt}(e^{-tQ} D e^{-tQ})$. Hence

$$C = \int_0^\infty e^{-tQ} D e^{-tQ}\, dt, \qquad 1 = c(v,v) = \sum_u \delta_u \int_0^\infty (e^{-tQ})^2_{vu}\, dt.$$

Summing over v and using the symmetry of e^{-tQ},

$$n = \sum_u \delta_u \int_0^\infty (e^{-2tQ})_{uu}\, dt = \frac{1}{2}\sum_u \delta_u (Q^{-1})_{uu}. \qquad \square$$

3. The diagonal of $(I+P)^{-1}$

Lemma 3.1. *For every vertex u,*

$$(Q^{-1})_{uu} \geq \frac{2n-3}{2(n-2)},$$

with equality if and only if u is adjacent to every other vertex.

Proof. Write $e_u = a + b$ with $a = \mathbf{1}/n$ and $b = e_u - \mathbf{1}/n \perp \mathbf{1}$. Since $\mathbf{1}$ is an eigenvector of Q with eigenvalue 2, the cross terms vanish and

$$(Q^{-1})_{uu} = a^\top Q^{-1} a + b^\top Q^{-1} b = \frac{1}{2n} + b^\top Q^{-1} b.$$

By the Cauchy–Schwarz inequality applied to $Q^{1/2}b$ and $Q^{-1/2}b$, $(b^\top b)^2 \leq (b^\top Q b)(b^\top Q^{-1} b)$, with equality if and only if Qb is a multiple of b. Here $b^\top b = 1 - \frac{1}{n}$, and since $P(u,u) = 0$,

$$b^\top Q b = e_u^\top Q e_u - a^\top Q a = 1 - \frac{2}{n} > 0.$$

Therefore

$$(Q^{-1})_{uu} \geq \frac{1}{2n} + \frac{(n-1)^2}{n(n-2)} = \frac{2n-3}{2(n-2)}.$$

For the case of equality, $Pb = \frac{1}{d}\mathbf{1}_{N(u)} - \frac{1}{n}\mathbf{1}$, where $N(u)$ is the neighbourhood of u. If $Pb = \mu b$, then the coordinate u gives $-\frac{1}{n} = \mu(1 - \frac{1}{n})$, so $\mu = -\frac{1}{n-1}$, while a coordinate $w \notin N(u) \cup \{u\}$ would give $-\frac{1}{n} = -\frac{\mu}{n}$, so $\mu = 1$. Hence equality forces $N(u) = V \setminus \{u\}$. Conversely, if u is adjacent to all other vertices then $d = n-1$, $G = K_n$, and one checks directly that $Pb = -\frac{1}{n-1} b$. $\square$

Using only $(Q^{-1})_{uu} \geq 1/Q_{uu} = 1$ in place of Lemma 3.1, the argument below gives $\operatorname{Var} S \leq n/2$, which is Proposition 14.22 of [1] with a non-strict inequality. The gain comes from treating the constant vector, on which Q acts by 2 rather than by something near 1, separately.

4. Proofs

Proof of Theorem 1.1. Put $m^* = \frac{2n-3}{2(n-2)}$. By Lemmas 2.2 and 3.1, and $\delta_u \geq 0$,

$$2n = \sum_u \delta_u m_u \geq m^* \sum_u \delta_u = 4m^* \operatorname{Var} S,$$

so $\operatorname{Var} S \leq n/(2m^*) = n(n-2)/(2n-3)$. If $G \neq K_n$, then G, being regular, has no vertex adjacent to all others, so $m_u > m^*$ for every u by Lemma 3.1; and $\sum_u \delta_u > 0$ by Lemma 2.2. Hence the inequality is strict. If $G = K_n$, every m_u equals m^* and equality holds, in agreement with (1.1). $\square$

Proof of Corollary 1.2. The linear system of Lemma 2.1 (the off-diagonal entries of (2.1) together with $c(v,v) = 1$) has at most one solution. Indeed, the difference E of two solutions has zero diagonal and $QE + EQ = F$ diagonal, so by the proof of Lemma 2.2 $E = \int_0^\infty e^{-tQ} F e^{-tQ}\, dt$, and its diagonal is Mf, where f is the diagonal of F and $M = \int_0^\infty e^{-tQ} \circ e^{-tQ}\, dt$. By the Schur product theorem M is positive definite, so $f = 0$ and $E = 0$. Every automorphism σ of G preserves the antivoter dynamics, so $c(\sigma v, \sigma w)$ also solves the system. Hence c is automorphism-invariant,

and on a vertex-transitive graph both $\delta_u = \delta$ and $m_u = \frac{1}{n}\operatorname{tr} Q^{-1}$ are constant. Lemma 2.2 gives $\delta \operatorname{tr} Q^{-1} = 2n$, and Lemma 2.1 gives $\operatorname{Var} S = \frac{n}{4}\delta = \frac{n^2}{2\operatorname{tr} Q^{-1}}$, with $\operatorname{tr} Q^{-1} = \sum_i (1+\lambda_i)^{-1}$. □

5. Numerical check

The correlation equations of Lemma 2.1 form a linear system in the $\binom{n}{2}$ unknowns $c(v,w)$, $v \neq w$, which we solved exactly for every connected non-bipartite regular graph on at most seven vertices, excluding cycles (Table 1), and for 297 random connected non-bipartite regular graphs with $5 \le n \le 14$ and $3 \le d \le n-1$. In every case $\operatorname{Var} S \le n(n-2)/(2n-3)$ with equality exactly for K_n, and every diagonal entry of $(I+P)^{-1}$ was at least m^*, strictly so when $d < n-1$. The formula of Corollary 1.2 agrees with the exact solution to within $4 \cdot 10^{-15}$ on eight non-bipartite vertex-transitive graphs, among them the Petersen graph ($\operatorname{Var} S = 40/13$), $C_3 \times C_3$, two circulants and the cycle C_7 (where $\operatorname{Var} S = 1$).

n	graph	d	$\operatorname{Var} S$
4	K_4	3	1.6000
5	K_5	4	2.1429
6	K_6	5	2.6667
6	octahedron	4	2.4000
6	prism $C_3 \times K_2$	3	1.9459
7	K_7	6	3.1818
7	4-regular (two graphs)	4	2.7067, 2.4653

Table 1. Exact stationary variance of S on every connected non-bipartite regular graph with at most seven vertices, cycles excluded.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

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Data availability

No data were used. The code for Table 1 and for the random-graph check, with its output, is available at `https://github.com/yspennstate/antivoter-variance-complete-graph`.

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